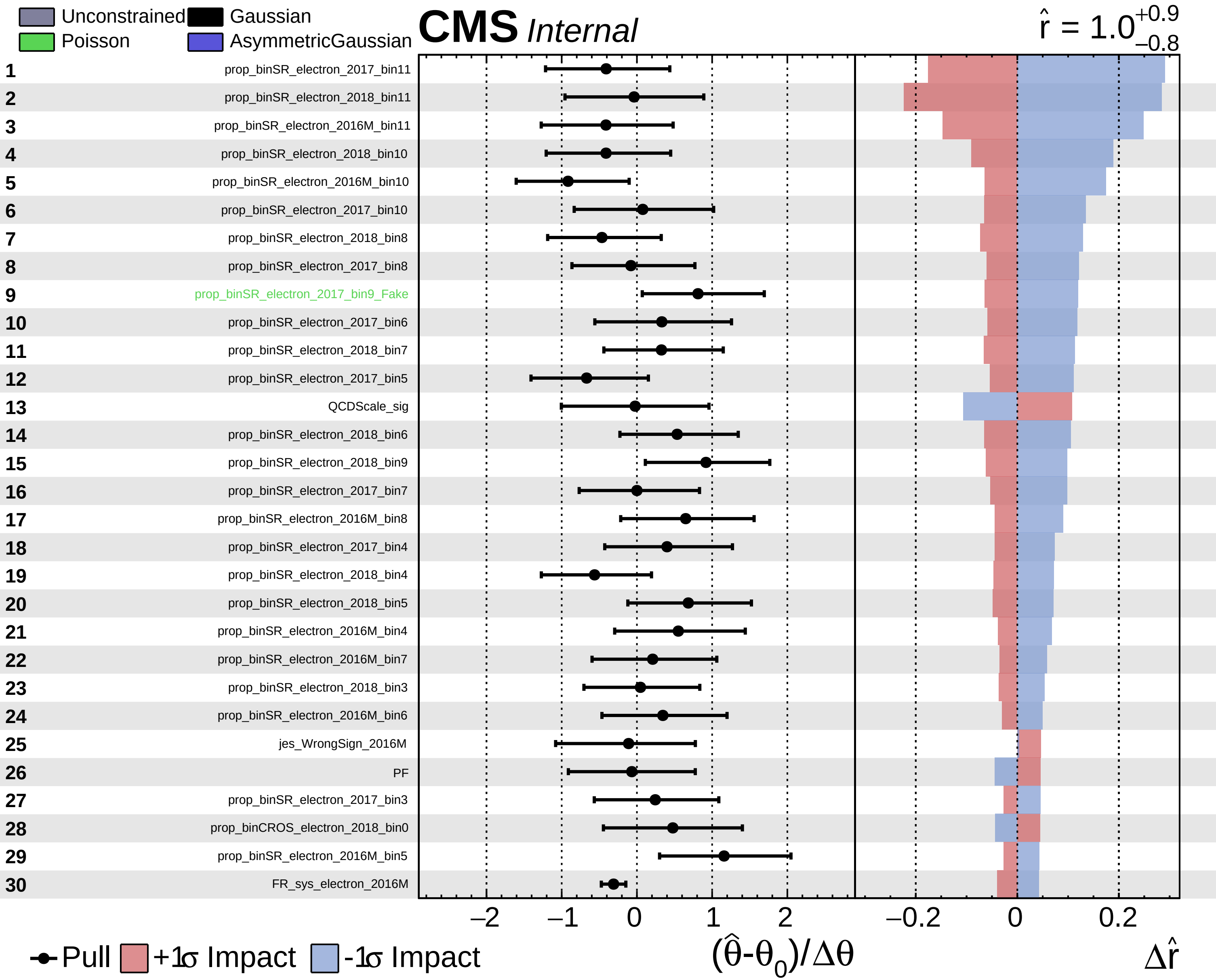
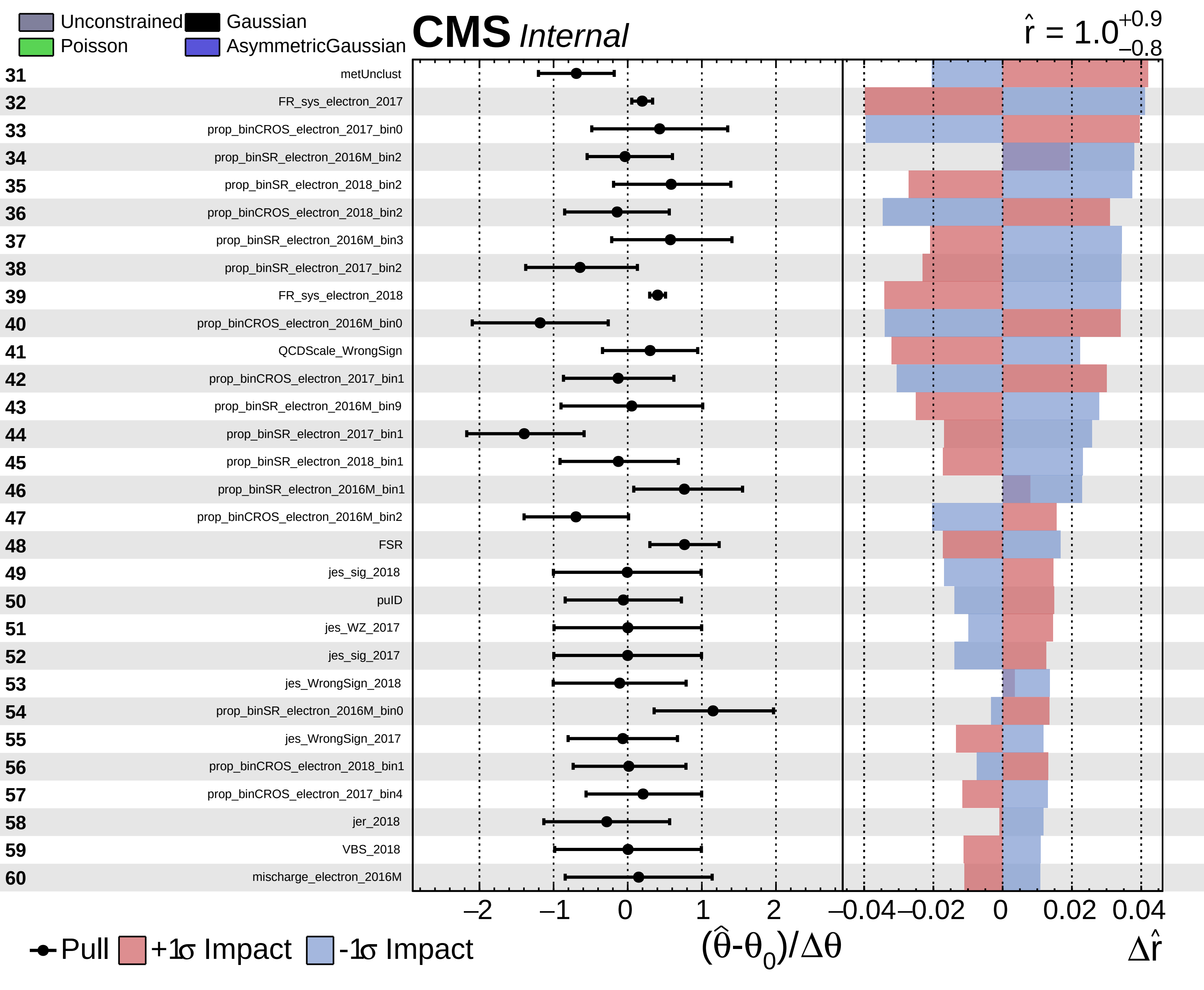
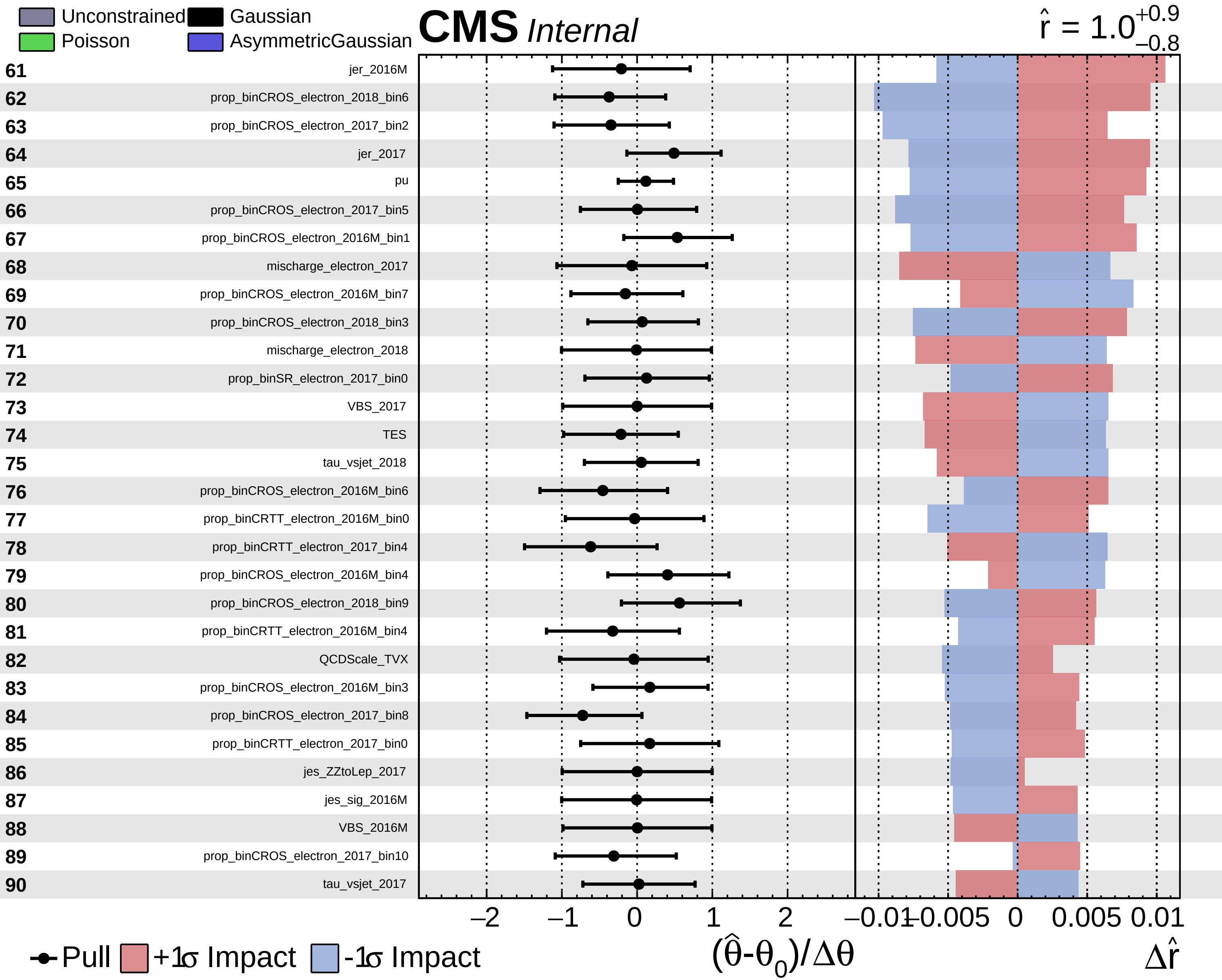
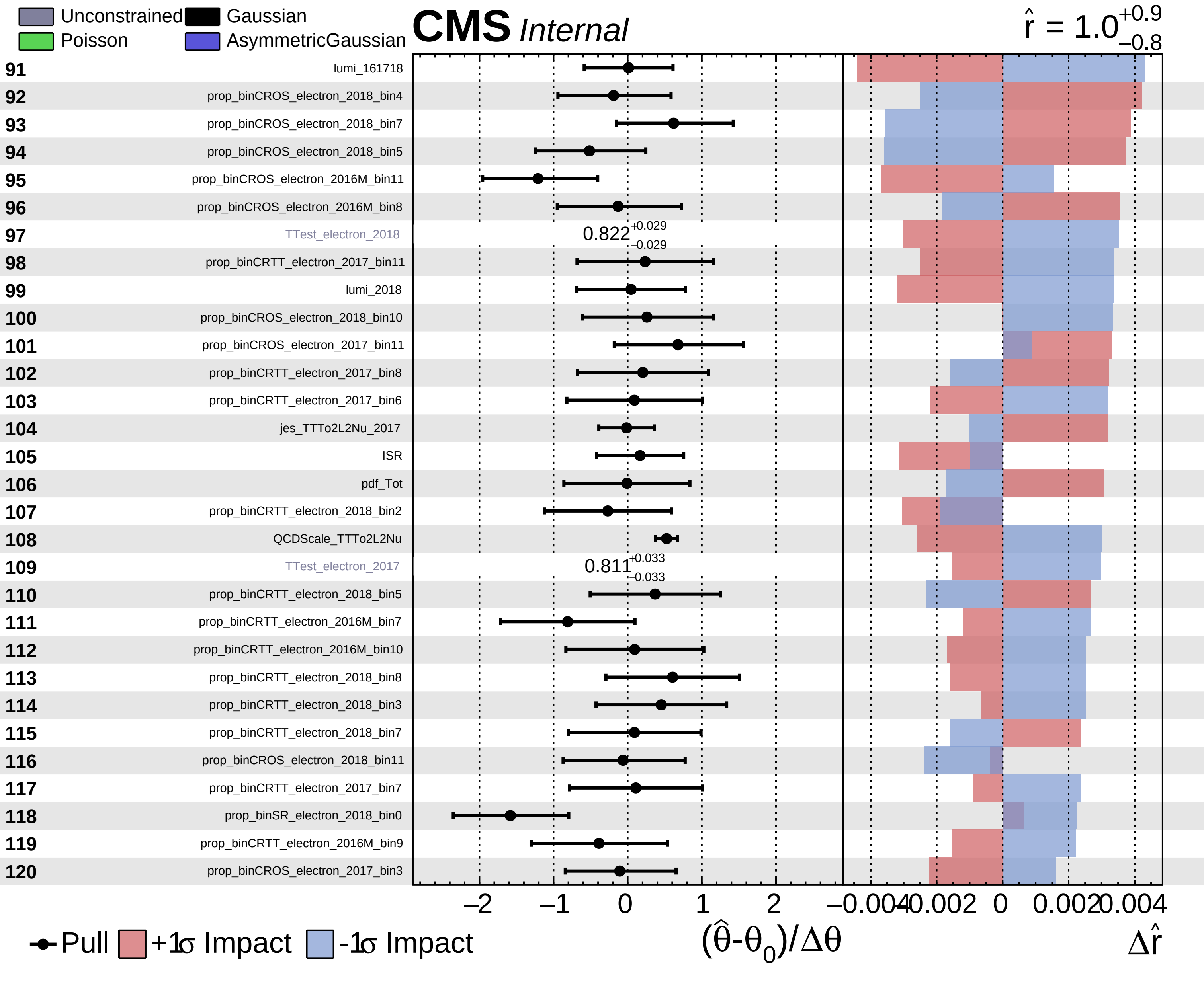


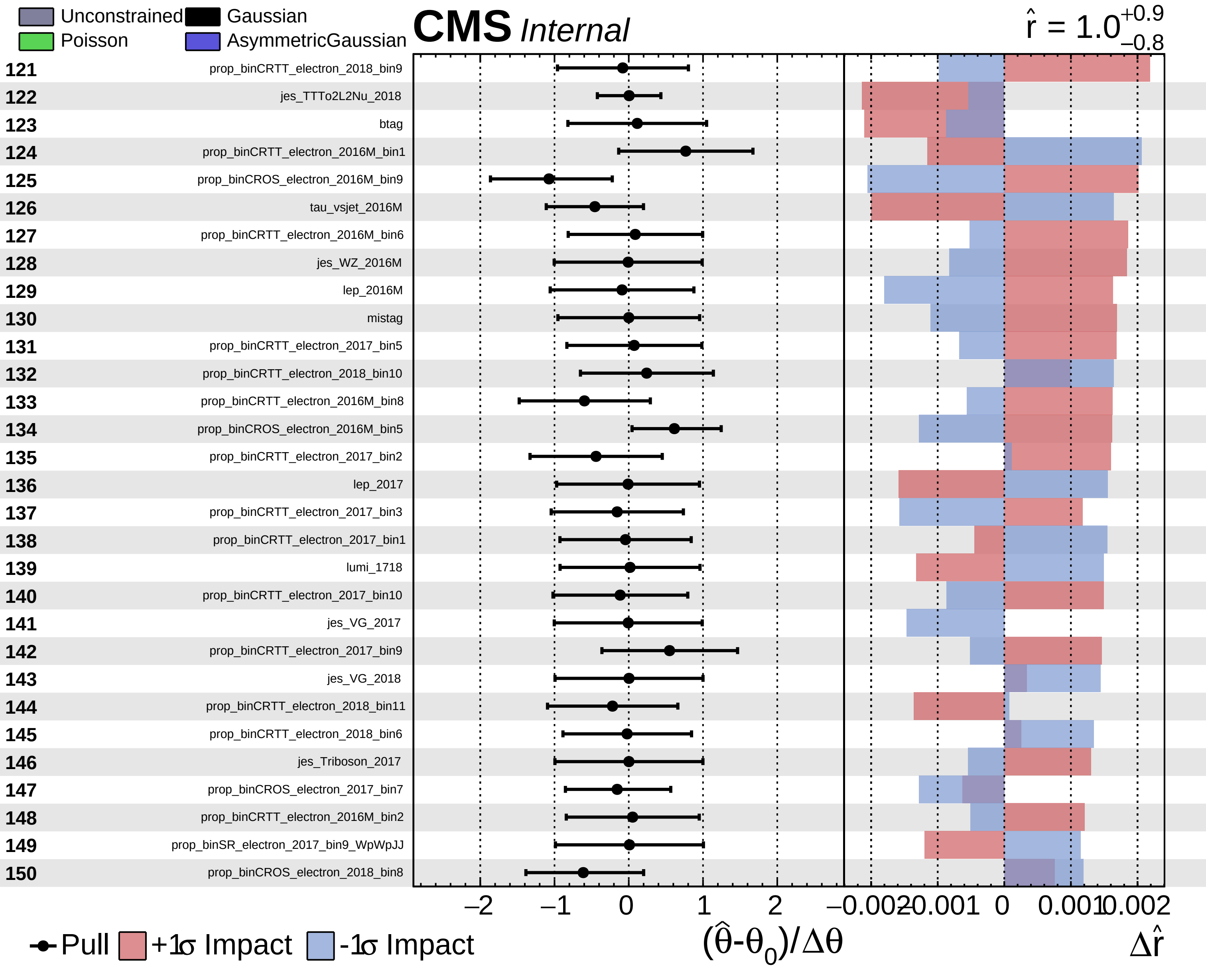
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
1 prop_binSR_electron_2017_bin11
2 prop_binSR_electron_2018_bin11
3 prop_binSR_electron_2016M_bin11
4 prop_binSR_electron_2018_bin10
5 prop_binSR_electron_2016M_bin10
6 prop_binSR_electron_2017_bin10
7 prop_binSR_electron_2018_bin8
8 prop_binSR_electron_2017_bin8
9 prop_binSR_electron_2017_bin9_Fake
10 prop_binSR_electron_2017_bin6
11 prop_binSR_electron_2018_bin7
12 prop_binSR_electron_2017_bin5
13 QCDScale_sig
14 prop_binSR_electron_2018_bin6
15 prop_binSR_electron_2018_bin9
16 prop_binSR_electron_2017_bin7
17 prop_binSR_electron_2016M_bin8
18 prop_binSR_electron_2017_bin4
19 prop_binSR_electron_2018_bin4
20 prop_binSR_electron_2018_bin5
21 prop_binSR_electron_2016M_bin4
22 prop_binSR_electron_2016M_bin7
23 prop_binSR_electron_2018_bin3
24 prop_binSR_electron_2016M_bin6
25 jes_WrongSign_2016M
26 PF
27 prop_binSR_electron_2017_bin3
28 prop_binCROS_electron_2018_bin0
29 prop_binSR_electron_2016M_bin5
30 FR_sys_electron_2016M
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
31 metUnclust
32 FR_sys_electron_2017
33 prop_binCROS_electron_2017_bin0
34 prop_binSR_electron_2016M_bin2
35 prop_binSR_electron_2018_bin2
36 prop_binCROS_electron_2018_bin2
37 prop_binSR_electron_2016M_bin3
38 prop_binSR_electron_2017_bin2
39 FR_sys_electron_2018
40 prop_binCROS_electron_2016M_bin0
41 QCDScale_WrongSign
42 prop_binCROS_electron_2017_bin1
43 prop_binSR_electron_2016M_bin9
44 prop_binSR_electron_2017_bin1
45 prop_binSR_electron_2018_bin1
46 prop_binSR_electron_2016M_bin1
47 prop_binCROS_electron_2016M_bin2
48 FSR
49 jes_sig_2018
50 puID
51 jes_WZ_2017
52 jes_sig_2017
53 jes_WrongSign_2018
54 prop_binSR_electron_2016M_bin0
55 jes_WrongSign_2017
56 prop_binCROS_electron_2018_bin1
57 prop_binCROS_electron_2017_bin4
58 jer_2018
59 VBS_2018
60 mischarge_electron_2016M
−2
−1
0
1
2
−0.04
−0.02
0
0.02
0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

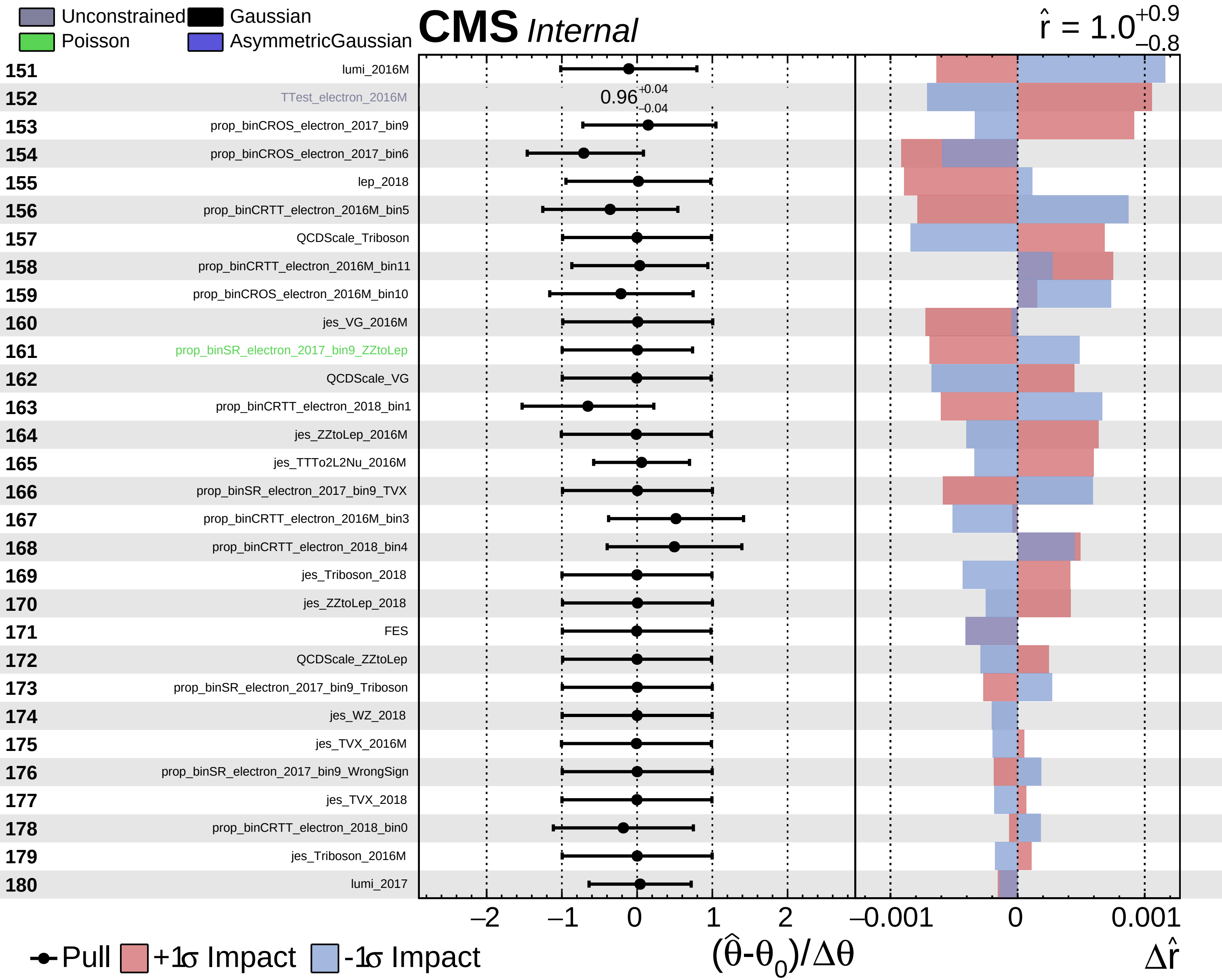

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
61 jer_2016M
62 prop_binCROS_electron_2018_bin6
63 prop_binCROS_electron_2017_bin2
64 jer_2017
65 pu
66 prop_binCROS_electron_2017_bin5
67 prop_binCROS_electron_2016M_bin1
68 mischarge_electron_2017
69 prop_binCROS_electron_2016M_bin7
70 prop_binCROS_electron_2018_bin3
71 mischarge_electron_2018
72 prop_binSR_electron_2017_bin0
73 VBS_2017
74 TES
75 tau_vsjet_2018
76 prop_binCROS_electron_2016M_bin6
77 prop_binCRTT_electron_2016M_bin0
78 prop_binCRTT_electron_2017_bin4
79 prop_binCROS_electron_2016M_bin4
80 prop_binCROS_electron_2018_bin9
81 prop_binCRTT_electron_2016M_bin4
82 QCDScale_TVX
83 prop_binCROS_electron_2016M_bin3
84 prop_binCROS_electron_2017_bin8
85 prop_binCRTT_electron_2017_bin0
86 jes_ZZtoLep_2017
87 jes_sig_2016M
88 VBS_2016M
89 prop_binCROS_electron_2017_bin10
90 tau_vsjet_2017
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



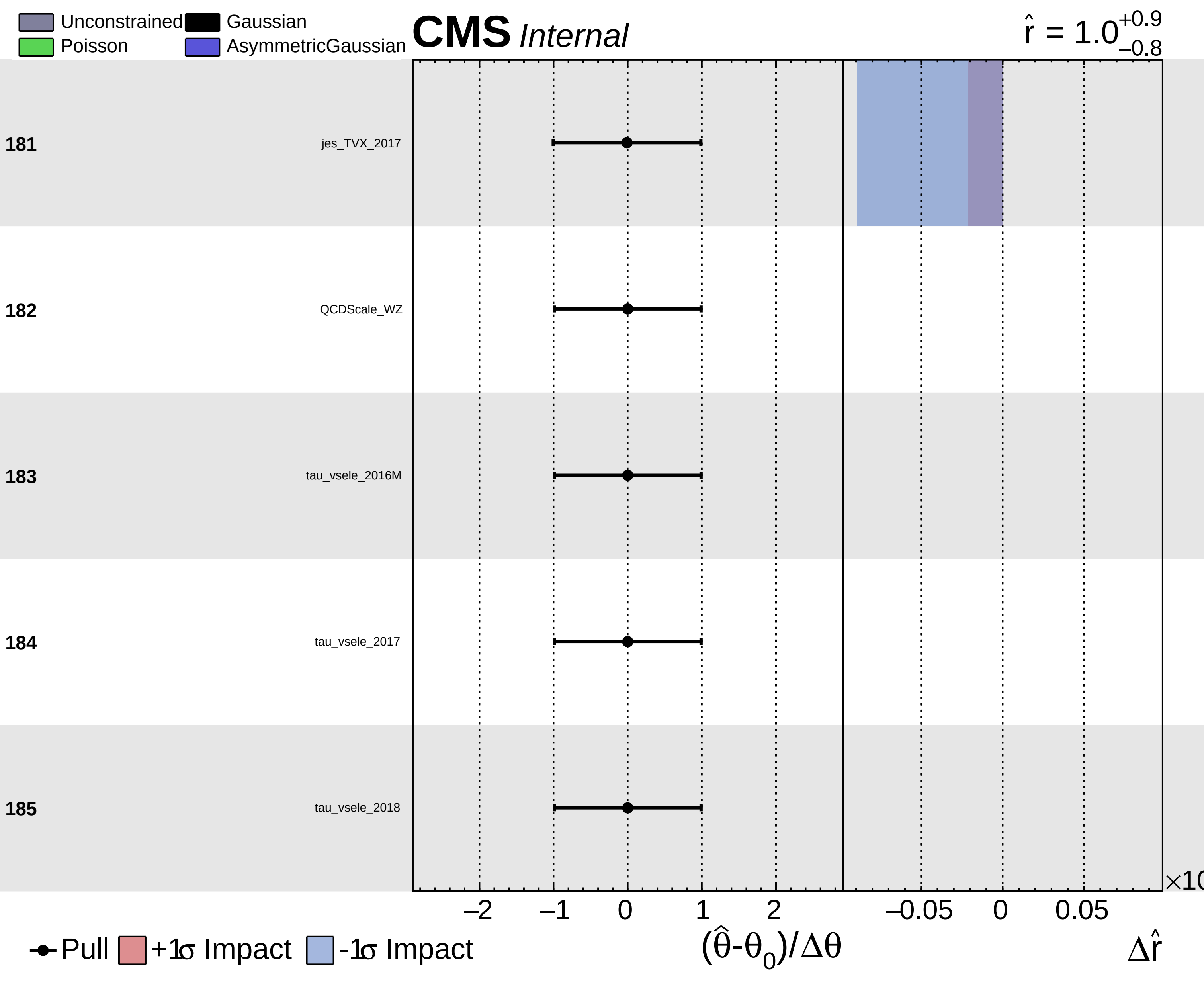
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
91 lumi_161718
92 prop_binCROS_electron_2018_bin4
93 prop_binCROS_electron_2018_bin7
94 prop_binCROS_electron_2018_bin5
95 prop_binCROS_electron_2016M_bin11
96 prop_binCROS_electron_2016M_bin8
97 TTest_electron_2018
$0.822^{+0.029}_{-0.029}$
98 prop_binCRTT_electron_2017_bin11
99 lumi_2018
100 prop_binCROS_electron_2018_bin10
101 prop_binCROS_electron_2017_bin11
102 prop_binCRTT_electron_2017_bin8
103 prop_binCRTT_electron_2017_bin6
104 jes_TTTo2L2Nu_2017
105 ISR
106 pdf_Tot
107 prop_binCRTT_electron_2018_bin2
108 QCDScale_TTTo2L2Nu
109 TTest_electron_2017
$0.811^{+0.033}_{-0.033}$
110 prop_binCRTT_electron_2018_bin5
111 prop_binCRTT_electron_2016M_bin7
112 prop_binCRTT_electron_2016M_bin10
113 prop_binCRTT_electron_2018_bin8
114 prop_binCRTT_electron_2018_bin3
115 prop_binCRTT_electron_2018_bin7
116 prop_binCROS_electron_2018_bin11
117 prop_binCRTT_electron_2017_bin7
118 prop_binSR_electron_2018_bin0
119 prop_binCRTT_electron_2016M_bin9
120 prop_binCROS_electron_2017_bin3
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
121 prop_binCRTT_electron_2018_bin9
122 jes_TTTo2L2Nu_2018
123 btag
124 prop_binCRTT_electron_2016M_bin1
125 prop_binCROS_electron_2016M_bin9
126 tau_vsjet_2016M
127 prop_binCRTT_electron_2016M_bin6
128 jes_WZ_2016M
129 lep_2016M
130 mistag
131 prop_binCRTT_electron_2017_bin5
132 prop_binCRTT_electron_2018_bin10
133 prop_binCRTT_electron_2016M_bin8
134 prop_binCROS_electron_2016M_bin5
135 prop_binCRTT_electron_2017_bin2
136 lep_2017
137 prop_binCRTT_electron_2017_bin3
138 prop_binCRTT_electron_2017_bin1
139 lumi_1718
140 prop_binCRTT_electron_2017_bin10
141 jes_VG_2017
142 prop_binCRTT_electron_2017_bin9
143 jes_VG_2018
144 prop_binCRTT_electron_2018_bin11
145 prop_binCRTT_electron_2018_bin6
146 jes_Triboson_2017
147 prop_binCROS_electron_2017_bin7
148 prop_binCRTT_electron_2016M_bin2
149 prop_binSR_electron_2017_bin9_WpWpJJ
150 prop_binCROS_electron_2018_bin8
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.9 −0.8
151 lumi_2016M
152 TTest_electron_2016M
0.96 +0.04 −0.04
153 prop_binCROS_electron_2017_bin9
154 prop_binCROS_electron_2017_bin6
155 lep_2018
156 prop_binCRTT_electron_2016M_bin5
157 QCDScale_Triboson
158 prop_binCRTT_electron_2016M_bin11
159 prop_binCROS_electron_2016M_bin10
160 jes_VG_2016M
161 prop_binSR_electron_2017_bin9_ZZtoLep
162 QCDScale_VG
163 prop_binCRTT_electron_2018_bin1
164 jes_ZZtoLep_2016M
165 jes_TTTo2L2Nu_2016M
166 prop_binSR_electron_2017_bin9_TVX
167 prop_binCRTT_electron_2016M_bin3
168 prop_binCRTT_electron_2018_bin4
169 jes_Triboson_2018
170 jes_ZZtoLep_2018
171 FES
172 QCDScale_ZZtoLep
173 prop_binSR_electron_2017_bin9_Triboson
174 jes_WZ_2018
175 jes_TVX_2016M
176 prop_binSR_electron_2017_bin9_WrongSign
177 jes_TVX_2018
178 prop_binCRTT_electron_2018_bin0
179 jes_Triboson_2016M
180 lumi_2017
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.9}_{-0.8}$
181 jes_TVX_2017
182 QCDScale_WZ
183 tau_vsele_2016M
184 tau_vsele_2017
185 tau_vsele_2018
−2 −1 0 1 2
−0.05 0 0.05
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$
Pull
+1σ Impact
-1σ Impact